SET 2430 Assignment 4

Part A. Count Nodes in a Binary Tree Using Recurrence

You are given the root of a binary tree.

Requirements:

1. Write a recursive function to count the total number of nodes in the binary tree.
2. Write the recurrence relation for your solution.
3. Solve the recurrence for a full binary tree of height h.
4. Analyze time complexity.

eg:

Input tree:



Output:

5

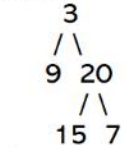
Part B.

Given the root of a binary tree, return its maximum depth.

The maximum depth is the number of nodes along the longest path from root to leaf.

Example:

Input:



Output:

3

Requirements:

1. Write recursive solution.
2. Write recurrence relation.
3. Analyze time complexity.